

ASSIGNMENT – 1

IMPLEMENT INHERITANCE TYPES USING JAVASCRIPT

Submitted By

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Subject: Full Stack Development

Objective

To implement and demonstrate different types of inheritance using JavaScript.

1. SINGLE INHERITANCE

Description: In single inheritance, one child class inherits from one parent class.

Output: The following screenshot shows the output obtained after executing the program.

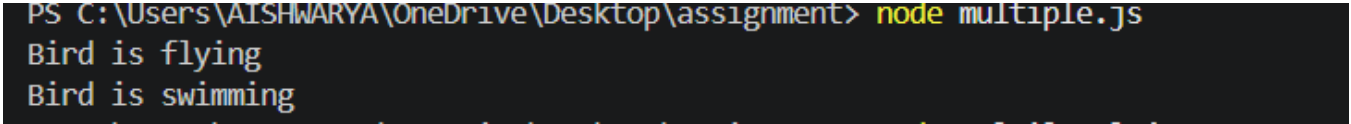
```
PS C:\Users\AISHWARYA\OneDrive\Desktop\assignment> node single.js
Animal is eating
Dog is barking
```

Program file: single_inheritance.js

2. MULTIPLE INHERITANCE

Description: JavaScript does not directly support multiple inheritance through classes. Mixins can be used to provide multiple sets of methods to a class.

Output: The following screenshot shows the output obtained after executing the program.

A screenshot of a terminal window with a black background and white text. The prompt is 'PS C:\Users\AISHWARYA\OneDrive\Desktop\assignment>' followed by the command 'node multiple.js'. The output consists of two lines: 'Bird is flying' and 'Bird is swimming'.

```
PS C:\Users\AISHWARYA\OneDrive\Desktop\assignment> node multiple.js
Bird is flying
Bird is swimming
```

Program file: multiple_inheritance.js

3. MULTILEVEL INHERITANCE

Description: In multilevel inheritance, a class inherits from another child class, forming a chain of inheritance.

Output: The following screenshot shows the output obtained after executing the program.

```
PS C:\Users\AISHWARYA\OneDrive\Desktop\assignment> node multilevel.js
Animal is eating
Dog is barking
Puppy is playing
```

Program file: multilevel_inheritance.js

4. HIERARCHICAL INHERITANCE

Description: In hierarchical inheritance, multiple child classes inherit from the same parent class.

Output: The following screenshot shows the output obtained after executing the program.

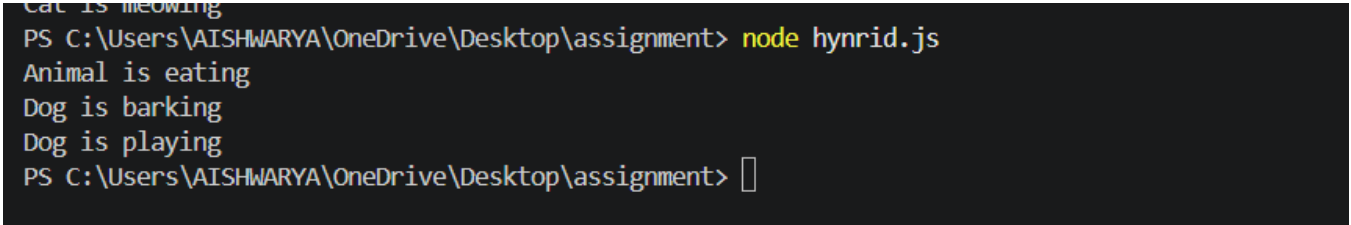
```
Puppy is playing
PS C:\Users\ATISHWARYA\OneDrive\Desktop\assignment> node hierarchical.js
Animal is eating
Dog is barking
Animal is eating
Cat is meowing
```

Program file: hierarchical_inheritance.js

5. HYBRID INHERITANCE

Description: Hybrid inheritance combines two or more inheritance techniques. In this program, class inheritance is combined with a mixin.

Output: The following screenshot shows the output obtained after executing the program.



```
Cat is meowing  
PS C:\Users\AISHWARYA\OneDrive\Desktop\assignment> node hynrid.js  
Animal is eating  
Dog is barking  
Dog is playing  
PS C:\Users\AISHWARYA\OneDrive\Desktop\assignment> 
```

Program file: hybrid_inheritance.js

CONCLUSION

The different types of inheritance were successfully demonstrated using JavaScript. The assignment covered Single, Multiple using Mixins, Multilevel, Hierarchical, and Hybrid inheritance. The programs were executed in the VS Code terminal and their outputs were recorded using screenshots.